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# Characterization of heavy metal resistance bacteria from Yamuna River and check the heavy metal tolerance activity of bacteria

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**Abstract-** The Yamuna River, revered as a holy river, faces severe pollution due to human activities, including waste dumping and industrial discharge. This contamination has rendered its water unsafe, raising significant concerns about pathogenic bacteria that can cause various diseases. This study collected water samples from the Kalindikunj barrage in Delhi NCR to isolate bacteria using LB agar media. The isolates were exposed to heavy metals, including Arsenic, Nickel Chloride, and Stannous Chloride, to assess their tolerance levels. Results indicated that bacteria from sample S3 exhibited maximum tolerance to Arsenic at 0.1 mM and minimum at 0.3 mM, while similar patterns were observed for other metals. Sample S4 showed maximum tolerance to Arsenic and other heavy metals at 0.1 mM, with varying minimum tolerances. These findings suggest that the identified heavy metal-tolerant bacteria could be valuable for bioremediation efforts to restore the Yamuna River's health.

**Key words: Yamuna River, pathogenic bacteria, heavy metal-tolerant bacteria, bioremediation**

## INTRODUCTION

The Yamuna River, one of India's most polluted water bodies, is severely affected by the accumulation of toxic heavy metals.[1] These metals, which naturally occur in the Earth's crust, are often introduced into the river system through domestic and industrial waste discharges. Common pollutants include chromium (Cr), mercury (Hg), lead (Pb), cobalt (Co), nickel (Ni), and cadmium (Cd).[2] While some metals like zinc and iron are essential for biological functions, others, such as arsenic and lead, pose significant health risks to humans and aquatic life.[3,4] Industrial practices that utilize heavy metals in manufacturing processes often lead to improper disposal, further exacerbating the pollution of water bodies.[5] These pollutants can spread through soil erosion and surface runoff during rainfall, affecting a broader ecological area.[6] The Yamuna River serves as a crucial water source for New Delhi but has increasingly become a dumping ground for industrial waste, leading to alarming levels of heavy metal contamination.

Heavy metals do not degrade easily and accumulate in sediments, posing long-term risks to ecosystems and human health. They disrupt aquatic food webs, affecting microbial diversity and potentially harming higher trophic levels through bioaccumulation.[7] Human exposure to these contaminants is linked to severe health issues, including cancer and organ damage. Moreover, heavy metals impact microbial communities by disrupting cell integrity, affecting

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their ecological roles.[8] Given these challenges, monitoring and addressing heavy metal pollution in the Yamuna River is crucial for protecting both environmental health and public safety. Identifying heavy metal-tolerant bacteria may provide a promising avenue for bioremediation, offering a cost-effective solution to mitigate pollution.[9,10]

## MATERIALS & METHODS

### Sample Collection

Water samples from the Yamuna River were collected at two sites—Site A and Site B near the Kalindikunj barrage in Delhi NCR. Four samples were taken from each site, spaced 50 meters apart, using sterile plastic bottles. The samples were immediately stored in a refrigerator at 4°C to preserve their integrity.

### Media Preparation

For the isolation of heavy metal-resistant bacteria, LB agar media and LB broth were utilized. The LB agar was prepared by dissolving 5.0 g of yeast extract, 10.0 g of tryptone, 10.0 g of NaCl, and 15.0 g of agar in distilled water to a final volume of 1000 mL. The mixture was autoclaved at 121°C and 15 psi for 15 minutes, then cooled and stored at 4°C.

### Bacterial Isolation through Serial Dilution

To isolate bacteria, a stock solution was created by adding 1 mL of the water sample to 9 mL of distilled water. This was thoroughly mixed. Serial dilutions were made ($10^{-1}$, $10^{-2}$, $10^{-3}$, and $10^{-4}$) by transferring 1 mL of the previous dilution into 9 mL of distilled water. From each dilution, 150 µL was spread onto LB agar plates, which were then incubated inverted at 37°C for 24 hours. For pure culture isolation, a sterile loop was used to streak diluted microbial suspensions onto prepared media plates, followed by 24-hour incubation at 37°C.

### Determination of Heavy Metal Tolerance

To assess heavy metal tolerance, isolated strains were inoculated into LB broth supplemented with increasing concentrations (0.1 mM, 0.2 mM, 0.7 mM, 0.9 mM) of metals like Arsenic, Cobalt, Stannous Chloride, and Nickel Chloride. After autoclaving, 50 µL of bacterial culture was added to 15 mL of broth in test tubes, which were incubated for 24 hours at 37°C. The maximum tolerance concentration (MTC) was identified as the highest metal concentration supporting bacterial growth, with controls maintained in metal-free LB broth.

### Heavy Metal Degradation Measurement

Heavy metal concentrations in the collected water samples were analyzed using a flame photometer, calibrated with standard solutions of heavy metals at 300 mg/L. For analysis, 300 mL of each water sample was prepared and examined in the photometer. All experiments were conducted in triplicates to ensure accuracy and reliability.

To calculate the percentage for degradation of heavy metal, formula can be used.

$$\%\ \text{Heavy Metal degradation} = \frac{\text{Heavy metal utelized } (\frac{\text{mg}}{\text{L}})}{\text{Heavy metal added to L. B. Broth } (\frac{\text{Mg}}{\text{L}})} X\ 100$$

## RESULT & DISCUSSION

This study aimed to isolate and characterize heavy metal-resistant bacteria from polluted sites of the Yamuna River and assess their metal tolerance levels. From Site-A, 35 bacterial isolates were obtained, while 20 isolates were recovered from Site-B. The bacteria were identified through morphology and Gram staining, followed by screening for their maximum tolerance to Arsenic, Cobalt, Stannous Chloride, and Nickel Chloride.

At Site-A, bacteria exhibited the highest tolerance to Arsenic, Cobalt, Stannous Chloride, and Nickel Chloride at concentrations of 0.1mM, with minimum tolerance observed at higher concentrations (0.2–0.5mM). Similarly, Site-B isolates showed maximum tolerance at 0.1mM and decreased resistance at higher concentrations (0.2–0.9mM). All isolates were Gram-positive with diverse morphologies.

The results were consistent with earlier findings, which reported arsenic concentrations in coal samples and industrial effluents exceeding safe limits. The maximum tolerance concentration (MTC) values for Nickel in this study were comparable to those found in Ni-tolerant bacteria from other studies, such as *Acinetobacter* isolates from Torsa River, India.

The LB broth cultures from both sites changed from transparent to translucent, indicating bacterial growth and tolerance to metals, as depicted in Figures 1. Tables 1-4 show detailed morphological characterizations and tolerance levels of bacteria to the tested metals across both sites. These findings highlight the metal-resistant potential of bacterial isolates from polluted environments.



**Figure 1- (a) Pure culture isolated from Site -A.**



**Figure 1- (b) Pure culture isolated from Site -B**

**Table 1- Morphological characterization of the selected bacterial isolates from Site-A**

| Identification | Sample 1 | Sample 2 | Sample 3 | Sample 4 |
|---|---|---|---|---|
| Arrangement | Scattered | Scattered | Scattered | Scattered |
| Gram Staining | +ve | +ve | +ve | +ve |
| Shape | Mostly Cocci, few rod shaped | Mostly Cocci, few were rod shaped | Mostly Cocci, few were rod shaped in culture | Cocci |

**Table 2- Morphological characterization of the selected bacterial isolates from Site-B**

| Identification | Sample 1 | Sample 2 | Sample 3 | Sample 4 |
|---|---|---|---|---|
| Arrangement | Scattered, few in Chains and bunches. | Scattered few in chains. | Scattered, few in Chains. | Scattered, few in bunches. |
| Gram Staining | +ve | +ve | +ve | +ve |
| Shape | Mostly Cocci, few were rod shaped. | Equally Cocci and Rod shaped | Equally Cocci and Rod shaped | Cocci |

Microorganisms, especially bacteria, possess superior heavy metal tolerance and removal capabilities. In this study, samples from Site-A and Site-B were tested with 300mg/L of heavy metals (Arsenic, Cobalt, Stannous Chloride, and Nickel Chloride) in LB broth. From Site-A, sample S2C9 showed 29% degradation of Arsenic, S2C3 degraded 19% of Cobalt, S3C4 degraded 88% of Stannous Chloride, and S2C9 degraded 30% of Nickel Chloride. In Site-B, sample S4C2 showed 63% degradation of Arsenic, S4C2 degraded 18% of Cobalt, S3C2 degraded 90% of Stannous Chloride, and S2C3 degraded 46% of Nickel Chloride. The results, shown in Tables 5 and 6, indicate that Site-B samples contained more metal-tolerant bacteria. Despite these degradations, Yamuna River water remains unsuitable for human consumption due to heavy metal contamination.

**Table 3- Percentage of Heavy Metal degradation of Samples by using different heavy metals from Site-A**

| Bacterial Culture Name | Concentration of heavy metal added in L.B. Broth | Used concentration of Arsenic | % of Arsenic Degradation | Used concentration of Cobalt | % of Cobalt degradation | Used concentration of Stannous Chloride | % of Stannous Chloride degradation | Used Concentration of Nickel Chloride | % of Nickel Chloride Degradation |
|---|---|---|---|---|---|---|---|---|---|
| Site -A S2C3 | 300 mg/L | 83 mg/L | 28 % | 58 mg/L | 19% | 103 mg/L | 34% | 63 mg/L | 21% |
| Site -A S2C9 | 300 mg/L | 88 mg/L | 29% | 54 mg/L | 18% | 102 mg/L | 34% | 82 mg/L | 27% |
| Site -A S3C2 | 300 mg/L | 84 mg/L | 28% | 47 mg/L | 16% | 260 mg/L | 87% | 63 mg/L | 21% |
| Site -A S3C4 | 300 mg/L | 85 mg/L | 28% | 48 mg/L | 16% | 263 mg/L | 88% | 61 mg/L | 20% |
| Site -A S4C3 | 300 mg/L | 29 mg/L | 9% | 12 mg/L | 4% | 42 mg/L | 14% | 22 mg/L | 7% |
| Site -A S4C5 | 300 mg/L | 34 mg/L | 11% | 23 mg/L | 8% | 183 mg/L | 61% | 89 mg/L | 30% |

**Table 4- Percentage of Heavy Metal degradation of Samples by using different heavy metals from Site- B**

| Bacterial Culture Name | Concentration of heavy metal added in L.B. Broth | Used concentration of Arsenic | % of Arsenic degradation | Used concentration of Cobalt | % of Cobalt degradation | Used concentration of Stannous Chloride | % of Stannous Chloride degradation | Used Concentration of Nickel Chloride | % of Nickel Chloride degradation |
|---|---|---|---|---|---|---|---|---|---|
| Site-B S1C3 | 300 mg/L | 32 mg/L | 10% | 11 mg/L | 3.6% | 78 mg/L | 26% | 92 mg/L | 31% |
| Site -B S1C4 | 300 mg/L | 33 mg/L | 11% | 18 mg/L | 6% | 72 mg/L | 24% | 71 mg/L | 24% |
| Site-B S2C2 | 300 mg/L | 39 mg/L | 13% | 13 mg/L | 4% | 218 mg/L | 73% | 121 mg/L | 40% |
| Site-B S2C3 | 300 mg/L | 38 mg/L | 13% | 17 mg/L | 6% | 251 mg/L | 84% | 138 mg/L | 46% |
| Site-B S3C2 | 300 mg/L | 63 mg/L | 21% | 13 mg/L | 4% | 271 mg/L | 90% | 92 mg/L | 31% |
| Site -B S4C2 | 300 mg/L | 189 mg/L | 63% | 53 mg/L | 18% | 103 mg/L | 34% | 88 mg/L | 29% |

## CONCLUSION

The continuous increase in heavy metal levels in the Yamuna River has led to the accumulation of toxic metals, posing serious risks to humans and animals. This contamination forces bacteria to develop mechanisms to survive in high metal concentrations, making them potential tools for removing heavy metals from polluted waters. This study found that bacteria from Site-A could tolerate high concentrations of Arsenic, Cobalt, Stannous Chloride, and Nickel Chloride at 0.1mM. In Site-B, bacteria tolerated Arsenic, Cobalt, Stannous Chloride at 0.1mM and Nickel Chloride at 0.2mM. Water samples from Kalindikunj Barrage, Delhi NCR, revealed metal concentrations exceeding safe limits for drinking, agriculture, and industry. The Yamuna River is unsafe for human consumption, urging reduced toxin dumping into water bodies. These metal-tolerant bacteria can be crucial for bioremediation, offering a solution to reduce heavy metal pollution and safeguard future generations.

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